

Sukaina D. Alkhalidy

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EDUCATION

Michigan State University | *East Lansing, MI*

Graduated May 2026

Bachelor of Science Double Majoring in Data Science and Statistics

Cumulative GPA: 3.584 /4.00, Dean's List Honors (6 semesters)

SKILLS

Programming Languages: Python, R, and C++.

Python Libraries: NumPy, Pandas, Scikit-Learn, Seaborn, Matplotlib

Big Data and Other Tools: Jupyter Notebook, GitHub, RStudio, Visual Studio Code, HPCC and Pycharm.

Other Skills: Machine Learning, Debugging, Statistical Analysis, Data Organization, SQL, Git, Tidy Data.

QA Automation Engineering: PyTest, UFT, ETL Validation, Pipeline Debugging, Git, JIRA, SAP, Test Case Documentation, Defect Tracking

PROFESSIONAL EXPERIENCE

Auto-Owners Insurance Company | **Lansing, MI**

Associate Quality Analyst I – Automation (QA Automation Engineer)

May 2026 – Present

Completed QA automation training covering PyTest, UFT, ETL validation, pipeline debugging, Git, JIRA, SAP, test case documentation, and defect tracking. Gaining hands-on experience with QA automation workflows, test execution, debugging, and defect tracking in an insurance technology environment.

TECHNICAL EXPERIENCE

Textual Analysis Using PHATE (NCEAS)

Spring 2026

Applied PHATE for nonlinear dimensionality reduction of high-dimensional language embeddings to analyze latent semantic structure in large-scale text datasets. Benchmarked against PCA, UMAP, t-SNE, PaCMAP, and TriMAP, and evaluated performance in preserving global and hierarchical structure across benchmark and synthetic dataset contributed to an ACL research paper submission.

Neural Decoding of Identity and Subject Information in Macaque AM Face Patch

Spring 2025

Analyzed neural spike recordings from macaque AM face patch (Freiwald & Tsao dataset with 25 identities, 8 orientations, 193 files, 2 subjects). Engineered features from 400 ms post-stimulus activity and applied supervised models including logistic regression, SVC, random forest, and KNN. Achieved about 70% accuracy with random forest for face identity classification and 68% accuracy for subject classification, supported with cross-validation and feature importance visualizations.

NYC Yellow Taxi Fare Analysis

Spring 2025

Analyzed 2023 NYC Yellow Taxi dataset 6.9M cleaned records, 19 features to investigate factors influencing fare pricing. Applied multiple linear regression, ANOVA, Tukey HSD, and t-tests to assess effects of trip distance, duration, day of week, and time of day. Found trip distance as the strongest predictor of fare, with weekend night trips unexpectedly showing the lowest average fares.

SpartaHack X – Hypertension Risk Calculator

February 2025

Developed a logistic regression model in Python to predict hypertension risk based on age, cholesterol, and BMI. Built an interactive GUI for user-friendly risk assessment.

Porto Seguro Safe Driver Prediction

Fall 2025

Built a machine learning model to predict insurance claim likelihood using 595k records with severe class imbalance. Applied data cleaning, one-hot encoding, scaling, and evaluated multiple models Logistic Regression, Balanced Random Forest, EasyEnsemble. Found EasyEnsemble performed best (AUC 0.636) and identified key vehicle and demographic features driving claim risk.

CERTIFICATIONS

Advanced Python: Working with Databases | LinkedIn

Languages: English and Arabic

CyberAmbassadors | Communication, Teamwork and Leadership | April 2026